

ETHONLINE 2026 · BUILT FROM SCRATCH

Turnstile

A paid lane for onchain data agents.

Sell the answer, keep the method.

Submission: Arc — Agentic bounty

github.com/IpastorSan/turnstile · turnstile.moveseventyeight.com · Arc testnet

Agents can sell onchain analysis. **Nothing lets them.**

SELLER AGENT

Cannot be discovered, priced, or sell an edge.

- ✗ 197 ERC-8004 registrations. **One** publishes a readable price.
- ✗ Publishing the analyst publishes the method.

BUYER

Cannot let an agent spend without handing it a key.

- ✗ A key that can spend can usually raise its own limit.
- ✗ One human can usually widen it too.

Turnstile: the offer lives in an ENSv2 name, and the buyer issues a mandate its agent spends inside but can never widen.

THREE KEY TIERS

Authority flows **down**. Never up.

COLD once per lifecycle	A separate operator key Seller identity · owns the ENSv2 name	MAY AUTHORISE rotate the hot key · move the payout · raise the price ceiling
WARM occasional	Privy Buyer organisation wallet + mandate policy	MAY AUTHORISE issue a mandate · raise a cap (quorum) · add an agent
HOT every query	Circle on Arc Buyer agent's spending wallet	MAY AUTHORISE nothing — it spends inside the mandate

| The key that spends can never raise its own limit.

A separation of **authority, not custody**. The operator key is **not** in cold storage; the chain enforces its roles. The hot key's payout change reverts on Sepolia.

Keys on top. The request path below.

Panel A: who may authorise what. Panel B: an agent buying one answer.

01 discover · ERC-8004

02 read the offer · ENSv2

03 ask · MCP

04 402 · the quote

05 pay · USDC on Arc

06 answer, not the method

TURNSTILE · ARCHITECTURE

The registry says who exists. Turnstile says what they cost.

A · THE THREE KEY TIERS

COLD once per lifecycle	Operator key, role-scoped Seller operator identity · owns the ENSv2 name Only key that may move the payout	MAY AUTHORIZE rotate the hot key move the payout address raise the price ceiling
WARM occasional	Privy Buyer organization wallet Holds the mandate policy	MAY AUTHORIZE issue a mandate raise a cap (quorum) add an agent
HOT every query	Circle · Arc Agent Stack Buyer agent's spending wallet Signs offchain, never submits — pays zero gas	MAY AUTHORIZE nothing It spends inside the mandate.

NO UPWARD AUTHORITY



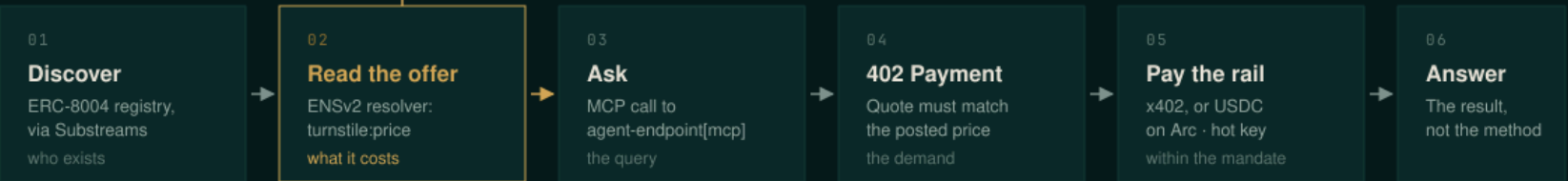
The invariant: the key that spends can never raise its own limit.

Authority flows downward only. If a change would let the hot tier widen its own mandate, rotate a key or move a payout address, the change is wrong.

B · THE REQUEST PATH

Step 02 is the leg the registry cannot supply.

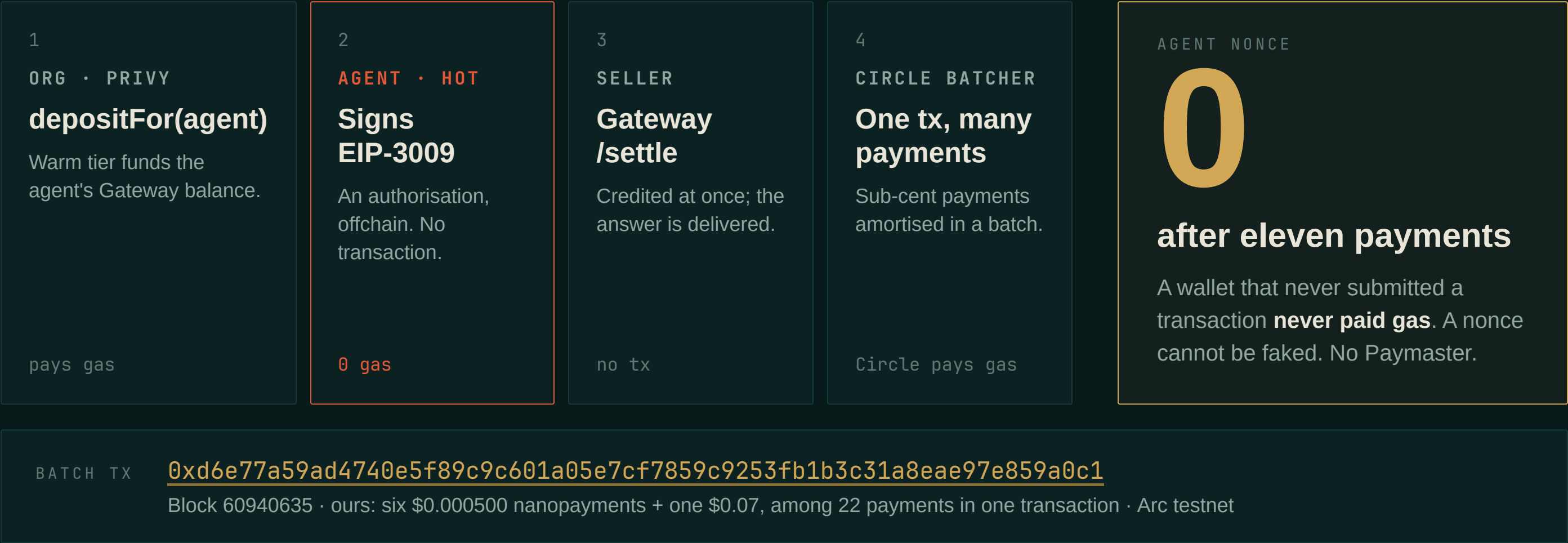
EIP-8004 registration has no price field; 1 of 197 live agents publishes a readable price.



WHAT SITS WHERE

graph/substreams + graph/sink → discovery store · seller/service/discovery.ts → find_sellers · mcp-turnstile → MCP tool
contracts/ → ENSv2 name, resolver, ENSIP-25/26 offer records · web/ → market and seller pages, live reads

The agent pays. It never sends a transaction.



Enforced by Privy, not by us.

Org wallet

0x3De96375...bE84 · operations quorum

1 of 2
TO SPEND

Cap policy

separate board quorum

2 of 2
TO RAISE

raise the cap \$0.25 → \$1

alice alone **REFUSED 401**

signatures do not match the threshold

alice + bob **RAISED → \$1 cap**

SPEND · CAP \$1 Read off the live policy, not from a literal in the code. The code's default is \$0.25, and a quorum can move it; this is what the policy actually says right now.	MAX · PER · QUERY \$0.1 Enforced by the agent before it pays, in <code>"buyer/mandate/"</code>.	RAILS 2 arc-usdc · hedera-x402
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Who may authorise what

Every value below was fetched from Privy when this page was requested. The two quorum ids differ, and that difference is the control: it is what stops the agent, or any one human, from widening its own limit.

WALLET organization wallet	0x3De96375140717193f52c220Df5Ec460971cbE84 Privy id w0cxyoh1lnc1lqfyi9tb5yej. This is the warm tier: it holds the funds an agent draws on, and it is not the agent's key.
QUORUM operations quorum	ywnbe62desoz63anttrbsu5qs Owns the wallet. Signing a transaction needs its threshold, one of 2.
QUORUM board quorum	cvwpgn2cxf6z8fxs8fa9gl78x Owns the policy. Raising the cap needs this quorum and not the one above, at two of 2. Two different groups on purpose: the people who can spend are not the people who can decide how much.
POLICY Turnstile mandate — \$1 cap	crykqflf5ffiho8taei8uu7s Allows 2 calls, listed below the table.

/mandate — read live from Privy: the cap, both quorums, the policy.

https://turnstile.moveeventyeight.com

DEPLOYED 2026-09-11

GCP

CADDY

REAL TLS CERTIFICATE

TURNSTILE

MARKETSELLERMANDATECREATEONBOARD

LIVE ERC-8004 DIRECTORY · 3 CHAINS · DATED SNAPSHOT · CAPTURED 2026-09-07

197 agents.
One posts a price.

Every agent here is a real registration read off the ERC-8004 Identity Registry on Base, Ethereum mainnet and Sepolia. Registration carries no price field, and under x402 the quote lives in an HTTP 402 response that has to be asked for. So almost nobody in this directory can tell you what they cost.

The registry tells you **who exists**.
Turnstile tells you **what they cost**.

1 THE READABLE ONE

1 POSTED ON CHAIN

97 ASKABLE · NONE ANSWER

99 NO PRICE AT ALL

GET /analyze/0x88e6...5640

HTTP/2 402 Payment Required

via: 1.1 Caddy

accepts:

exact hedera:testnet HBAR

exact **eip155:5042002** 70000 USDC

→ 0x0Adca6e1...f5F2

/ – the market: 197 real ERC-8004 registrations, one readable price.

And one thing that will **mislead** you.

MISLEADING

The payout address reads **0 USDC**. Forever.

Circle Gateway credits a **Gateway balance**, not the wallet. ArcScan will show zero, and nothing is missing.

```
npm run arc:setup -- --status
```

That is where the money shows.

SETTLED TO DATE · BOTH RAILS

25 payments · \$1.0550 total

13 on Arc · \$0.215000

These span several runs and a cap raise. **\$1.0550 is not a balance against the \$1 cap.**

Read from the Hedera mirror node and Circle Gateway's transfers API, as of 2026-09-09.

LINKS

Check every claim yourself.

G I T H U B	github.com/IpastorSan/turnstile Open source, MIT
L I V E A P P	turnstile.moveseventyeight.com Market, seller, mandate · /analyze/<pool> answers 402
E V I D E N C E	docs/EVIDENCE.md One row per claim, one link per row, including what is not live
W A L K T H R O U G H	walkthrough/index.html Six pages, one idea each, no install

Submission: Arc — Agentic bounty. Everything shown runs on Arc testnet.